Author Guidelines for ACCV Submission

Anonymous ACCV 2026 Submission

Paper ID #*****

1 ECCV review

reviewer1:

1. The core idea is a combination of SVD low-rank compression and entropy-based adaptation. The contribution of an adaptive compression strategy seems to be incremental, with not enough algorithmic contributions. **[nothing we can do]**
2. The comparisons and ablations are insufficient, as only plain SVD and W-SVD are included in the evaluation. Other acceleration/quantization compression pipelines are suggested for comparison. Also, the evaluation of combining this method with other tools is highly suggested. **[maybe we can add some experiments]**
3. The paper does not explain how to select the threshold-partitioning parameter α according to different datasets, tasks, or hardware platforms. **[we need to clarify that we always use 0.25 (did not tune) in the paper]**
4. In Table.9, the results of "original" and "diverse" are exactly the same. More check about the correctness and reason for that is suggested. **[we need to clarify that the learned thresholds did not actually change significantly]**

reviewer2:

1. the novelty is marginal, as long as the improvements in terms of results **[nothing we can do]**
2. The method itself has a lot of hyperparameters that have been fixed empirically without further justification. This can make the method not able to generalize well. For example the number of elements in the calibration set, the nature of the images, the number of the bins in the entropy estimation **[we need to clarify WHY pick certain values for these hyperparameters]**
3. What is the overall advantages of this method? it is not the memory footprint since you still need to store the entire model. The author should clarify better this point and introduce some quantitative analysis of this method's benefit. **[we need to clarify that we are not saving the WHOLE model; we are saving model of the highest retention regime, which is already much compressed!]**
4. qualitative examples are not really useful, I suggest to devote less space on them **[will adjust based on the paper length accordingly]**

reviewer3:

1. The proposed method leverages SVD low-rank representation of linear layers, which is a well-studied technique. The proposed method is incremental with respect to the state of the art. The major contribution is to make SVD rank selection adaptive with respect to input complexity. However, the proposed approach is highly empirical and may lack generality **[nothing we can do]**
2. The use of the first-order entropy of grayscale pixel intensities to rank input complexity may be oversimplistic since it neglects spatial correlation; more importantly, 3D complexity is not addressed at all. The ablation study in Sext. 4.4 only adds views (Otsu segmentation, Canny edge detection) but still relies on first-order entropy. **[we should clarify that our method is both simple and lightweight enough - we do not need a super precise 'difficulty estimator']** **[we may need to supply more experiments for 3D complexity]**
3. The experimental validation is sufficient, but solely based on comparison with two baselines (SVD and whitening SVD). The gain of the proposed adaptive SVD with 3 rank ratios over fixed rank baselines represents an expected results. **[we should clarify that even if the gain is 'expected', the methodology itself is still important/significant]**
4. Still, it is important to prove that the proposed method is competitive or can be integrated with the state of the art, e.g. the mentioned FastVGGT, LiteVGGT, FlashVGGT, QuantVGGT. It is not clear whether the proposed solution can be combined with a token merging or a quantization approach, yielding significant gains. **[we should emphasize our method is INDEED orthogonal - parameter pruning and token pruning are essentially independent]** **[we can add more experiments if we end up having time]**

Abstract. Recent feed-forward visual geometry models have demonstrated remarkable success in jointly predicting diverse geometric attributes — including depth, camera pose, and point clouds — within a single forward pass. Despite their strong performance, these models incur substantial computational cost: linear projections in both attention and MLP layers constitute the dominant bottleneck, accounting for the majority of FLOPs during inference. In this work, we revisit low-rank weight decomposition for visual geometry models via singular value decomposition (SVD), aiming to reduce computational complexity and accelerate inference. While SVD effectively reduces parameter count and FLOPs, we find that fixed-ratio compression is suboptimal, as the importance of singular values varies considerably across input samples. Motivated by this observation, we propose **SVD^3**, a training-free, data-adaptive compression framework for visual geometry models. SVD^3 introduces an entropy-guided compression allocation strategy that adaptively decomposes linear layers into low-rank matrices, and a retention mapping module that assigns each input a discrete retention ratio under a predefined compute budget estimated on a calibration dataset. Extensive experiments demonstrate that SVD^3 reduces inference cost by over 30%